

Wi-Fi router is up and running as soon as you switch it on, it's because someone probably worked very hard to ensure that it behaves flawlessly.

Designing hardware for embedded systems requires a deep understanding of the real-world scenarios within which electronic systems are used. If you're planning to embark on a full-fledged career in this field (more on this in a later chapter), this chapter will help you understand the issues you'll be up against. Even if you're not an electronics expert, we aim to help you understand how critical systems are designed. Let's start off with the heart and soul of any system – the processor.

Design challenges when making a processor

Power consumption – Why more is not always better

Like we already said, you're not going to find a Core i7 or Phenom in an embedded system. Why? Because typically embedded systems have processors that are very low power, which also means low complexity. A PC has enough power to feed these x86 monsters, but embedded systems don't.

An x86 (or x64) processor belongs to a kind of processor family known as CISC which stands for "Complex Instruction Set Computer". This type of processor has a more complicated instruction set as compared to another family of processors that's called RISC (Reduced Instruction Set Computer). It's RISC processors that are mainly used in embedded systems. A CISC processor is easier to control and command because it carries out some of the more complicated functions automatically, while a RISC processor isn't as smart. The programmer or compiler is tasked with the job of writing instructions for the processor to help it carry out more complicated jobs.

Again, you might wonder why bother with RISC processors, why not just use CISC processors all the time? This is because simpler RISC-based systems cannot afford to use as much power as a CISC system. How would you like it if your simple router used as much power as a PC? Besides, if we used CISC processors everywhere, everything would cost that much more and be about as big as a PC. This is why ARM processors are majorly used in mobile phones – ARM processors are built using the RISC instruction set architecture.

Word length – For how long can it think?

Let's assume an 'elevator control' system is our target system. How much data would the system need to process in single cycles? The largest number the system would have to deal with would be the highest number of floors the elevator has to travel up. Assuming that the number is 50 (for a 50-storey building), you'd